

Sushan Harish

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Full Right to Work in the UK • Immediately Available • Open to Contract / FTC / Interim

PROFESSIONAL SUMMARY

Aerospace Systems/Test Engineer with hands-on experience across the full test rig lifecycle — schematic design, subsystem integration (fluid, electrical, DAQ and control), instrumentation, commissioning, and verification & validation (V&V) prior to entry into service (EIS). Most recently developed and commissioned a Rolls-Royce fuel/oil systems environmental test rig at Novel Engineering Consultants, working across 5+ concurrent rig programmes for Rolls-Royce, Vertical Aerospace, and MBDA. Experienced in writing test procedures and technical reports, running safety/sealing investigations, and liaising with suppliers and multidisciplinary stakeholders. Background in AS9100/EASA Part 145 quality engineering adds a regulatory-grade approach to documentation and traceability. MSc Aerospace Engineering (Distinction), Kingston University. Immediately available.

KEY SKILLS FOR TEST RIG ENGINEERING

- **Test Rig Development & Commissioning:** full lifecycle from schematic design through to commissioning and EIS readiness
- **Verification & Validation (V&V):** repeatability testing, baseline comparison, performance validation, systematic troubleshooting
- **Subsystem Design & Integration:** Fluid Mechanical (hydraulic/oil circuits, pressure regulation), Electrical (wiring, controller integration), Data Acquisition & Control (PicoLogger, Arduino/stepper-motor automation)
- **Instrumentation:** flowmeters, pressure transducers, temperature probes, torque measurement (Magtrol) — installation, calibration and scaling verification
- **Safety & Risk:** sealing/leakage investigations, vacuum and high-temperature safety assessments across altitude test envelope (-0.2 to -1.0 bar)
- **Documentation:** test procedures, validation reports, measurement schedules, engineering handover documentation
- **Supplier & Stakeholder Coordination:** procurement liaison for hydraulic hoses, pressure chambers and high-temperature equipment; multidisciplinary team coordination
- **Quality & Compliance:** AS9100, EASA Part 145, FMEA, RCA/RCCA, SAP NetWeaver
- **Engineering Tools:** Siemens NX, ANSYS, MATLAB, CFD, Excel VBA

PROFESSIONAL EXPERIENCE

Systems Engineer (Contract) | Novel Engineering Consultants, UK

March 2026 – June 2026

- Development, integration, commissioning and validation of a Rolls-Royce fuel/oil systems environmental test rig, spanning 5+ concurrent rig programmes for Rolls-Royce, Vertical Aerospace, and MBDA.
- Ran verification and validation activities ahead of test entry: repeatability assessment, baseline comparison, pump/valve validation, and comparative performance testing between rig configurations (e.g. original vs. upgraded G-Rotor assemblies).
- Designed and commissioned environmental test capability across vacuum, altitude (five setpoints from -0.2 to -1.0 bar), freezer, and elevated-temperature test envelopes.
- Delivered subsystem integration across fluid mechanical (hydraulic/oil circuit, back-pressure and flow control), electrical (wiring, controller, heater and sensor integration) and DAQ/control (PicoLogger, Arduino-driven stepper motor automation) disciplines.
- Installed, calibrated and validated instrumentation including flowmeters, pressure transducers, temperature probes and Magtrol torque measurement, verifying scaling and measurement accuracy against PicoLogger data.
- Investigated and resolved sealing, leakage, vacuum-holding and pressure-instability issues as part of ongoing safety and reliability assurance.
- Authored test procedures, validation reports, measurement schedules and engineering handover documentation; liaised directly with suppliers on hydraulic hoses, pressure chambers and high-temperature equipment.
- Built Excel VBA automation converting hundreds of raw CSV test files into formatted engineering reports, reducing manual data-processing time and improving reporting consistency.

Aerospace Quality & MRO Engineer | Honeywell Aerospace, UK

April 2025 – January 2026

- Supported AS9100 and EASA Part 145 compliance, presenting internal audit findings ahead of EASA/UK CAA regulatory assessments.
- Managed Non-Conformance Report (NCR) and RCCA investigation reporting, translating defect findings into corrective action recommendations.
- Maintained component traceability and quality documentation in SAP NetWeaver, supporting audit-readiness.
- Inspected and assembled aircraft wheel and brake components to CMM/SPM standards.

Aerospace Engineering Intern | Hindustan Aeronautics Limited, India

2022

- Supported overhaul and manufacturing operations within an aerospace OEM environment; communicated inspection findings and corrective actions.
- Updated technical documentation and quality records to maintain compliance with internal and statutory requirements.

Precision Engineering Intern | Lohatec Precision Pvt Ltd, India

2022

- Performed dimensional inspections and reported tolerance conformity against engineering drawings to production teams.
- Assisted quality and process improvement activities, maintaining production records and component traceability.

Captain / Founder | SAE Aero Club, India

2020 – 2023

- Led a 100-person multidisciplinary team through the full UAV engineering lifecycle, including requirements definition, design, build and flight test.
- Developed flight-test protocols using iterative data modelling to optimise thrust-to-weight ratio and aerodynamic efficiency.
- Presented technical progress and design decisions to competition judges and external stakeholders at national SAE competitions.

EDUCATION

MSc Aerospace Engineering (Distinction)

Kingston University, London | 2024 – 2025 | Award: Best Postgraduate Project, School of Engineering

BEng Mechanical Engineering (Distinction)

St Joseph Engineering College, India | 2019 – 2023

CERTIFICATIONS

- FMEA Training | FMEA Specialist Certification (Accredited)
- Explore Engineering Job Simulation | GE Aerospace
- Root Cause Analysis and the 8D Problem Solving Bootcamp | Udemy